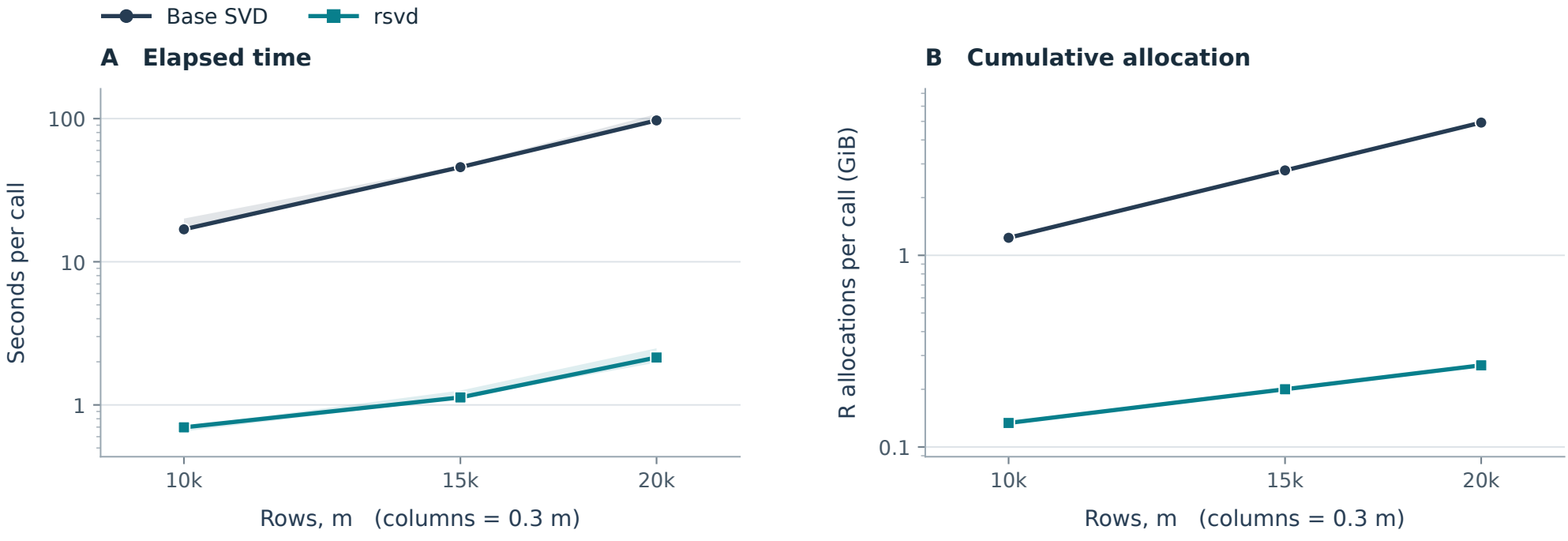
Dense SVD as matrix dimensions grow

30–120 million entries; geometric low-rank signal plus Gaussian noise



k = 40, p = 10, q = 2 · 5-run timing medians, IQR shaded · allocations: one profile · BLAS threads: 4

Largest observed error / optimal rank-40 error: 1.0007. R allocations exclude setup and do not measure peak RSS.